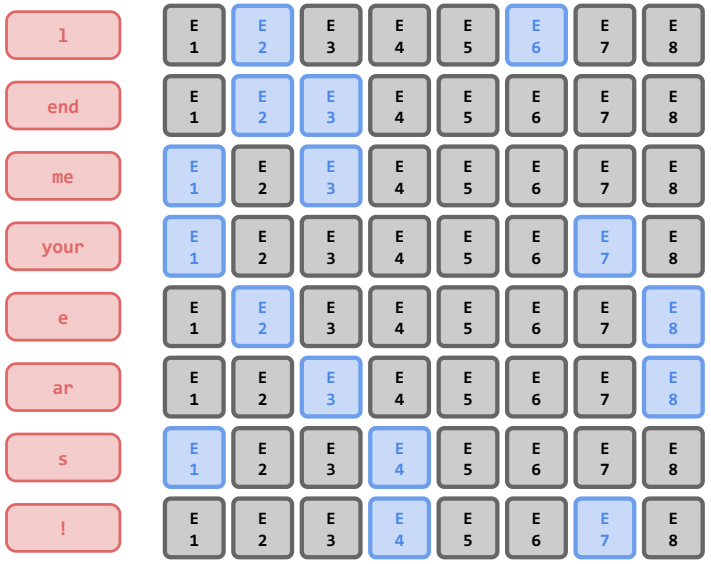
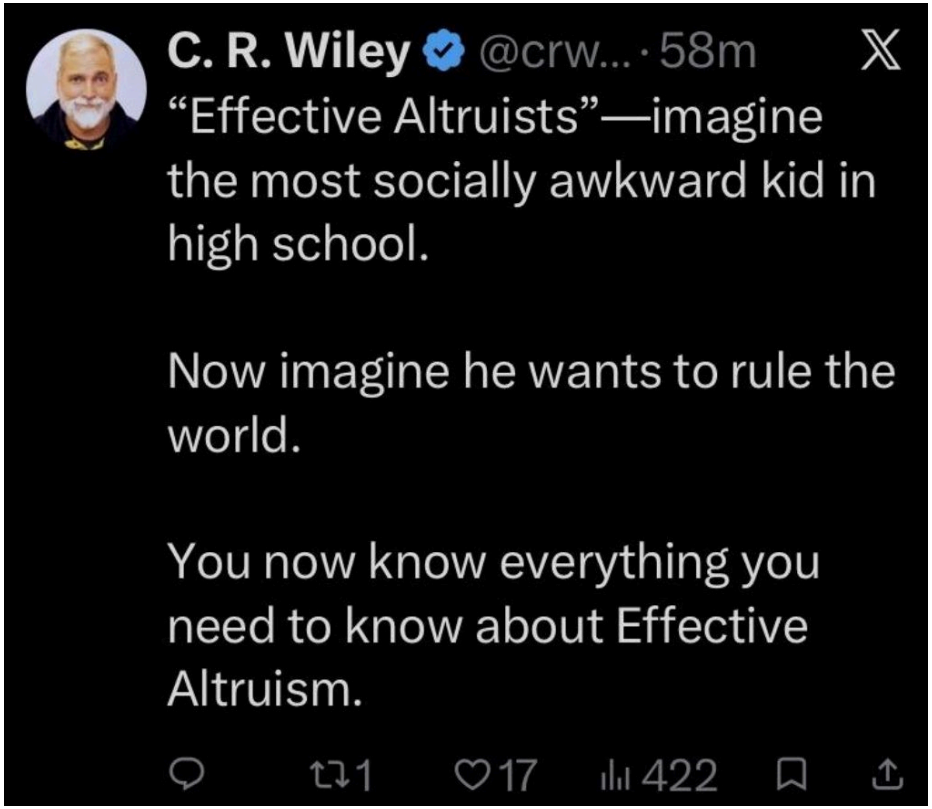
UNIT 01 · CHAPTER 11

Mixture of Experts

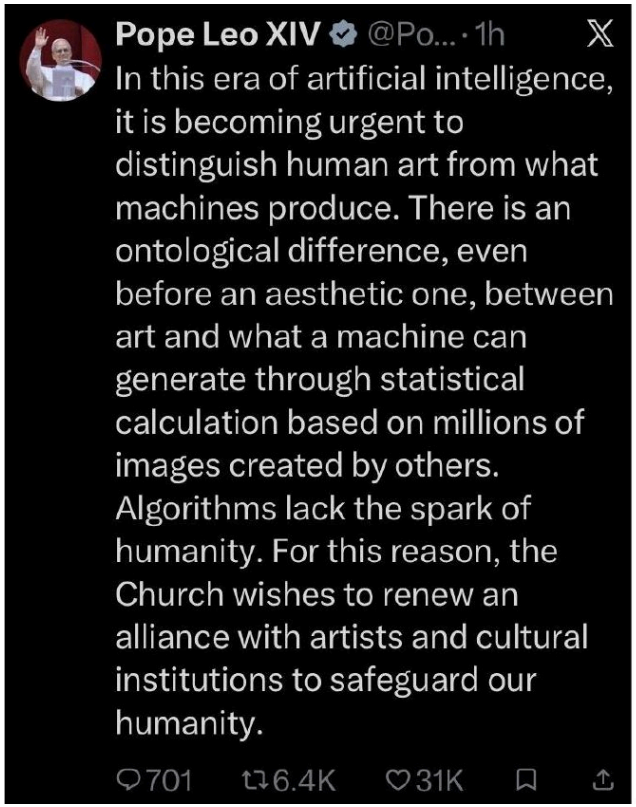
CSE 10124 · Building ChatGPT
William Theisen · University of Notre Dame



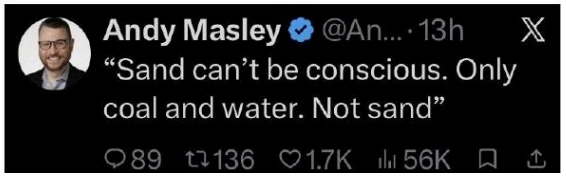
In the News


C. R. Wiley @crw... · 58m
“Effective Altruists”—imagine the most socially awkward kid in high school.
Now imagine he wants to rule the world.
You now know everything you need to know about Effective Altruism.
1
17
422

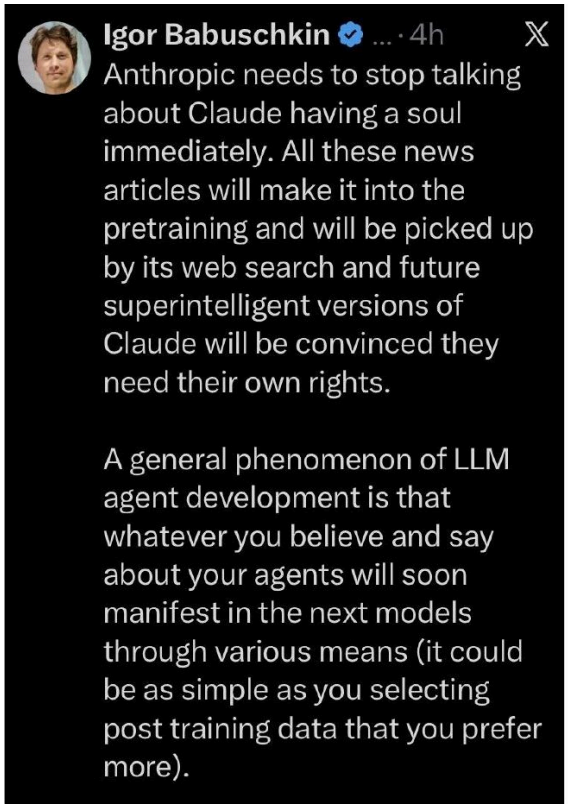
In the News

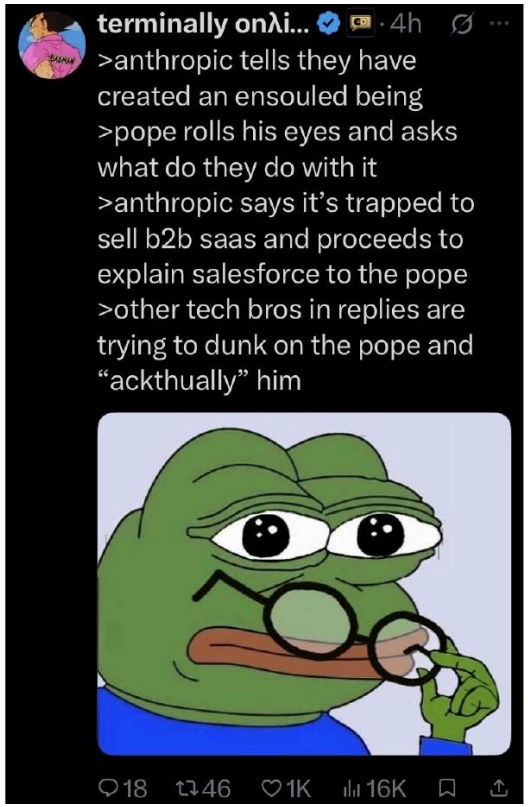

Pope Leo XIV @Po... · 1h
In this era of artificial intelligence, it is becoming urgent to distinguish human art from what machines produce. There is an ontological difference, even before an aesthetic one, between art and what a machine can generate through statistical calculation based on millions of images created by others. Algorithms lack the spark of humanity. For this reason, the Church wishes to renew an alliance with artists and cultural institutions to safeguard our humanity.
701
6.4K
31K


Christopher Hale · 3h
NEW: According to a bombshell report in the New York Times, Anthropic co-founder Chris Olah threatened to walk out of Pope Leo XIV's AI encyclical launch in May because the pope rejected the idea that machines can be conscious.
Olah's team then privately lobbied the pope's advisers "to take the possibility of model consciousness seriously." Pope Leo XIV held firm.

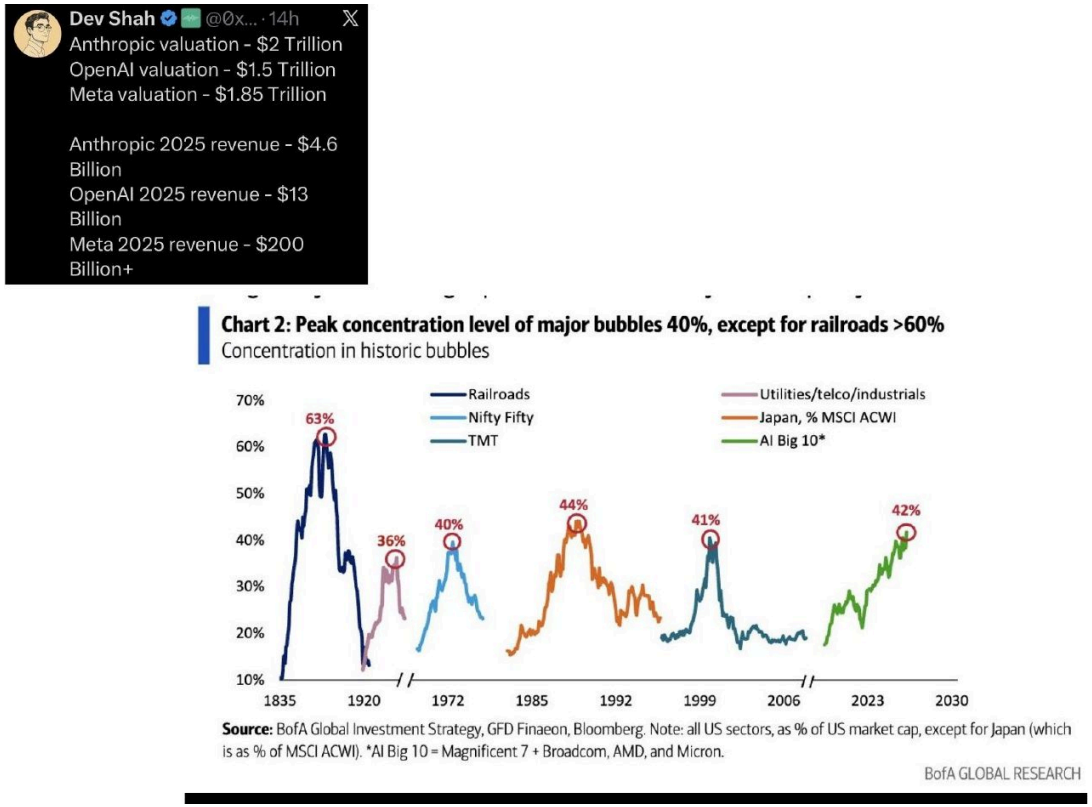

Andy Masley @An... · 13h
"Sand can't be conscious. Only coal and water. Not sand"
89
136
1.7K
56K

In the News

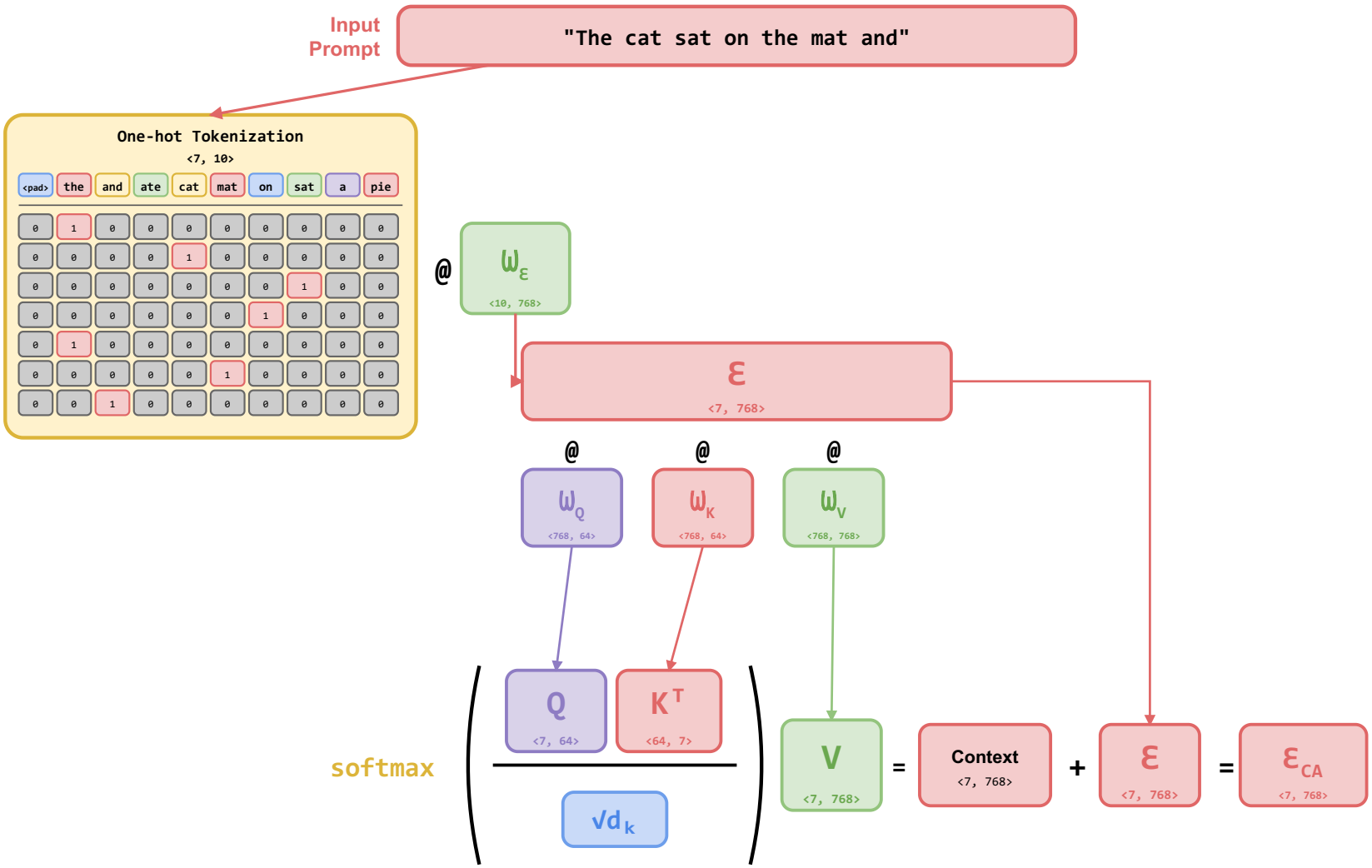

Igor Babuschkin ... · 4h
Anthropic needs to stop talking about Claude having a soul immediately. All these news articles will make it into the pretraining and will be picked up by its web search and future superintelligent versions of Claude will be convinced they need their own rights.
A general phenomenon of LLM agent development is that whatever you believe and say about your agents will soon manifest in the next models through various means (it could be as simple as you selecting post training data that you prefer more).


terminally onλi... · 4h
>anthropic tells they have created an ensouled being
>pope rolls his eyes and asks what do they do with it
>anthropic says it's trapped to sell b2b saas and proceeds to explain salesforce to the pope
>other tech bros in replies are trying to dunk on the pope and "ackthually" him
18
46
1K
16K

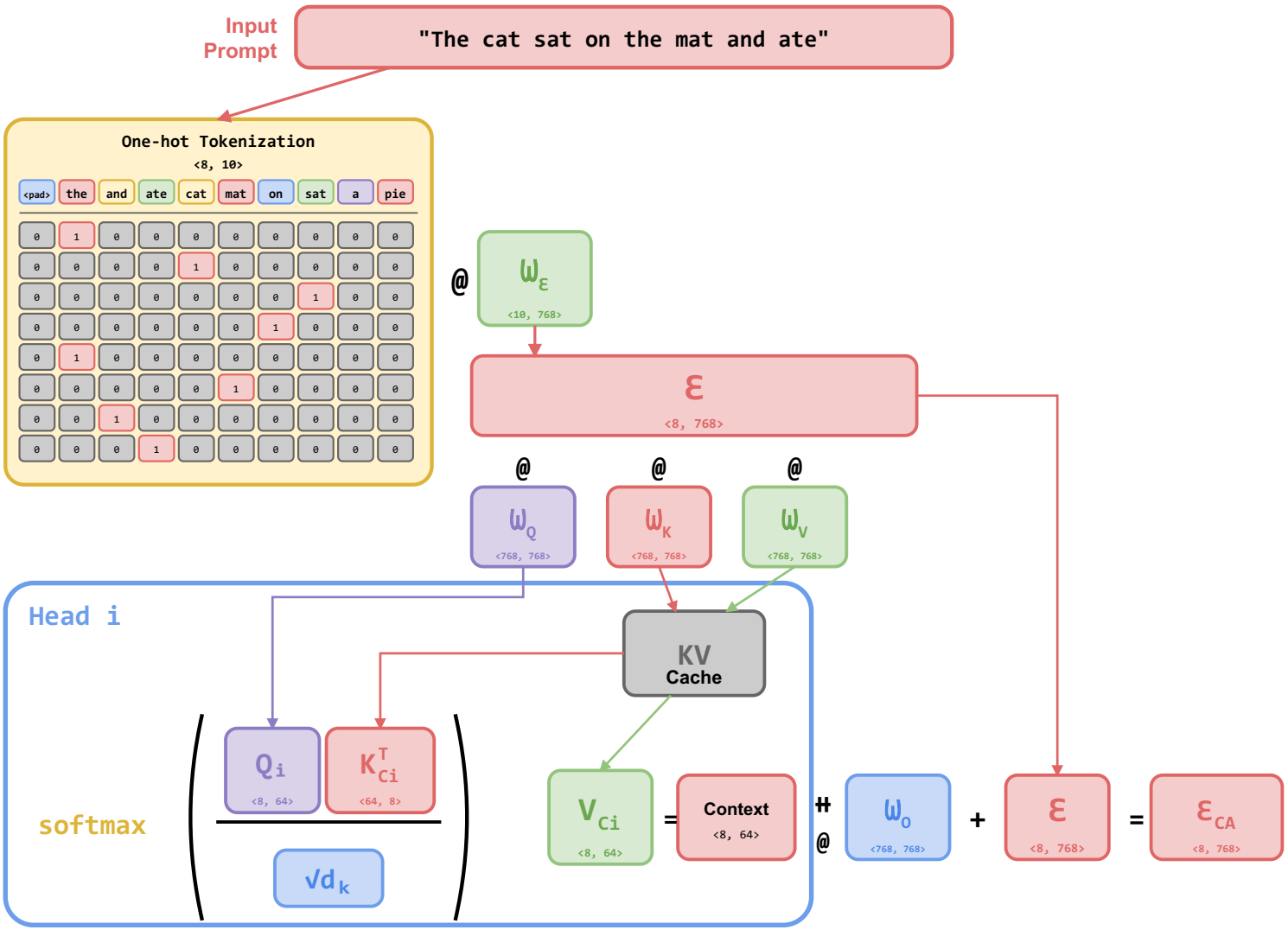
In the News


Dev Shah @0x... · 14h
Anthropic valuation - $2 Trillion
OpenAI valuation - $1.5 Trillion
Meta valuation - $1.85 Trillion
Anthropic 2025 revenue - $4.6 Billion
OpenAI 2025 revenue - $13 Billion
Meta 2025 revenue - $200 Billion+
Chart 2: Peak concentration level of major bubbles 40%, except for railroads >60%
Concentration in historic bubbles
Railroads
Nifty Fifty
TMT
Utilities/telco/industrials
Japan, % MSCI ACWI
AI Big 10*
70%
60%
50%
40%
30%
20%
10%
63%
36%
40%
44%
41%
42%
1835
1920
1972
1985
1992
1999
2006
2023
2030
Source: BofA Global Investment Strategy, GFD Finaeon, Bloomberg. Note: all US sectors, as % of US market cap, except for Japan (which is as % of MSCI ACWI). *AI Big 10 = Magnificent 7 + Broadcom, AMD, and Micron.
BofA GLOBAL RESEARCH

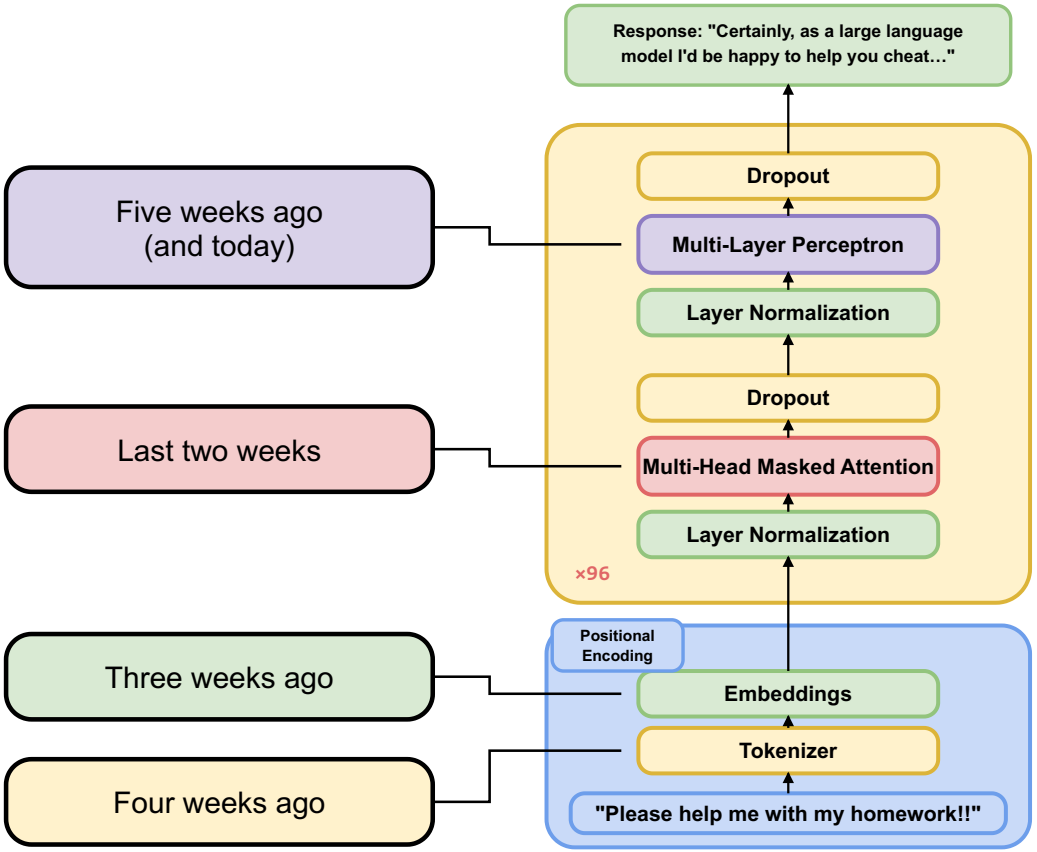
Standard Attention



Multi-Head Attention with KV Cache



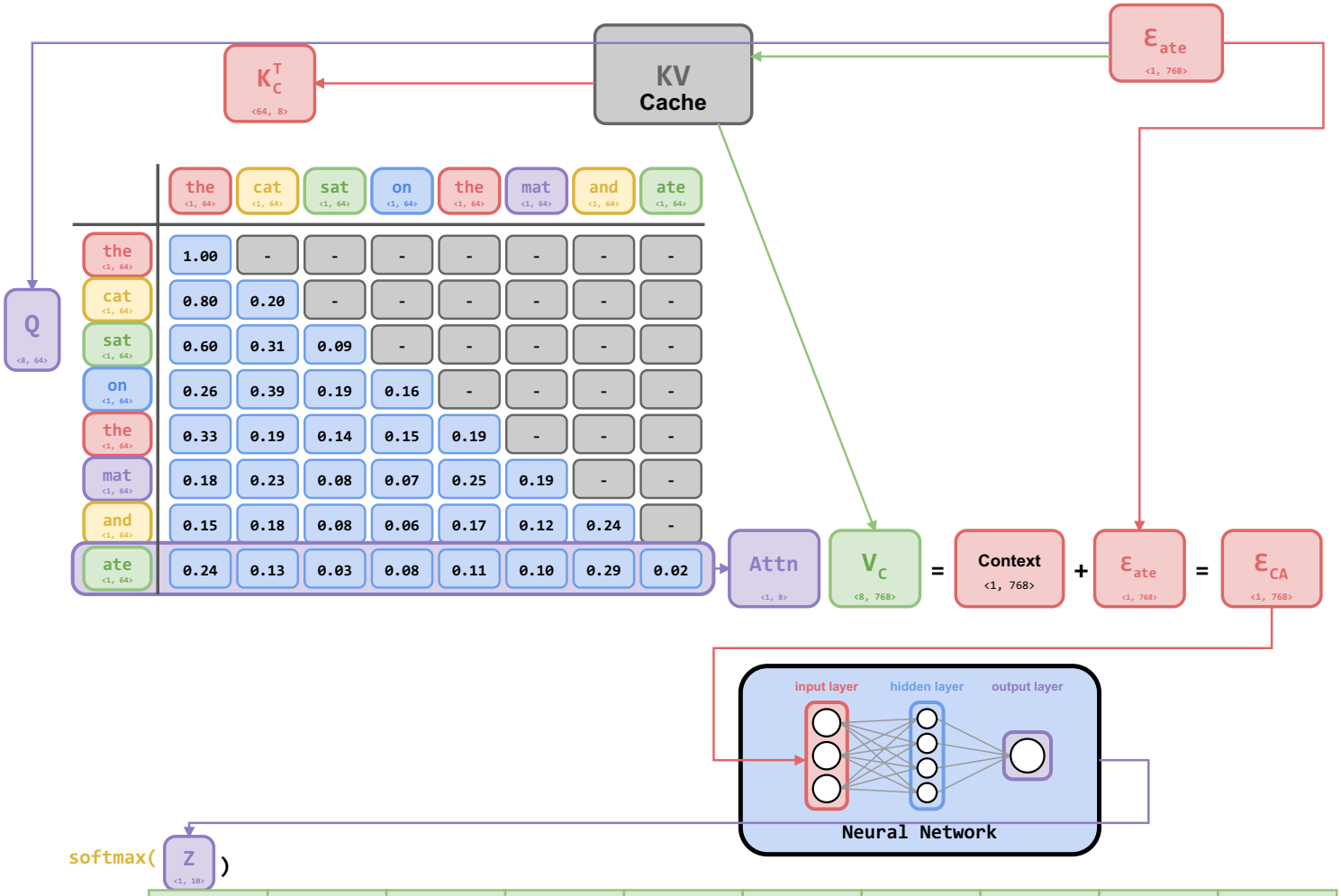
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Attention is the technique through which we produce stronger, context-aware embeddings for our input tokens.

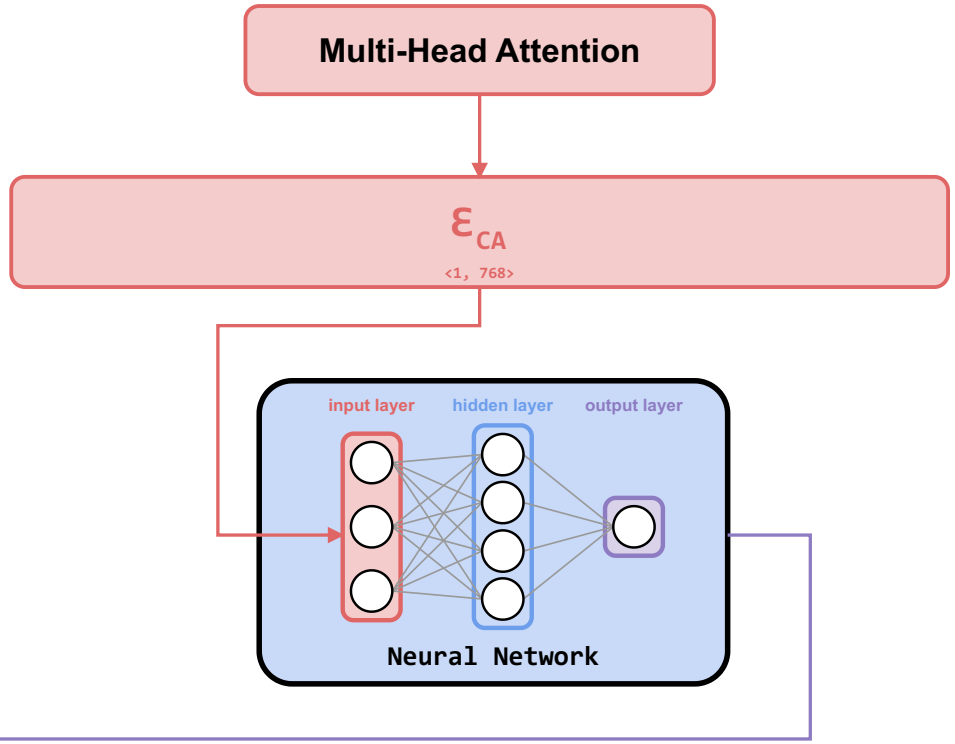
The MLP is where the majority of the knowledge and facts the LLM knows are stored and where we use the stronger embeddings.

Producing the output prediction



	0.1739	0.1079	0.0981	0.0779	0.0629	0.0614	0.0393	0.0241	0.0223	0.0116
Token ID	**262**	257	465	13	607	663	11	340	617	319
Token	**" the"**	"a"	" his"	"."	" her"	" its"	","	" it"	" some"	" on"

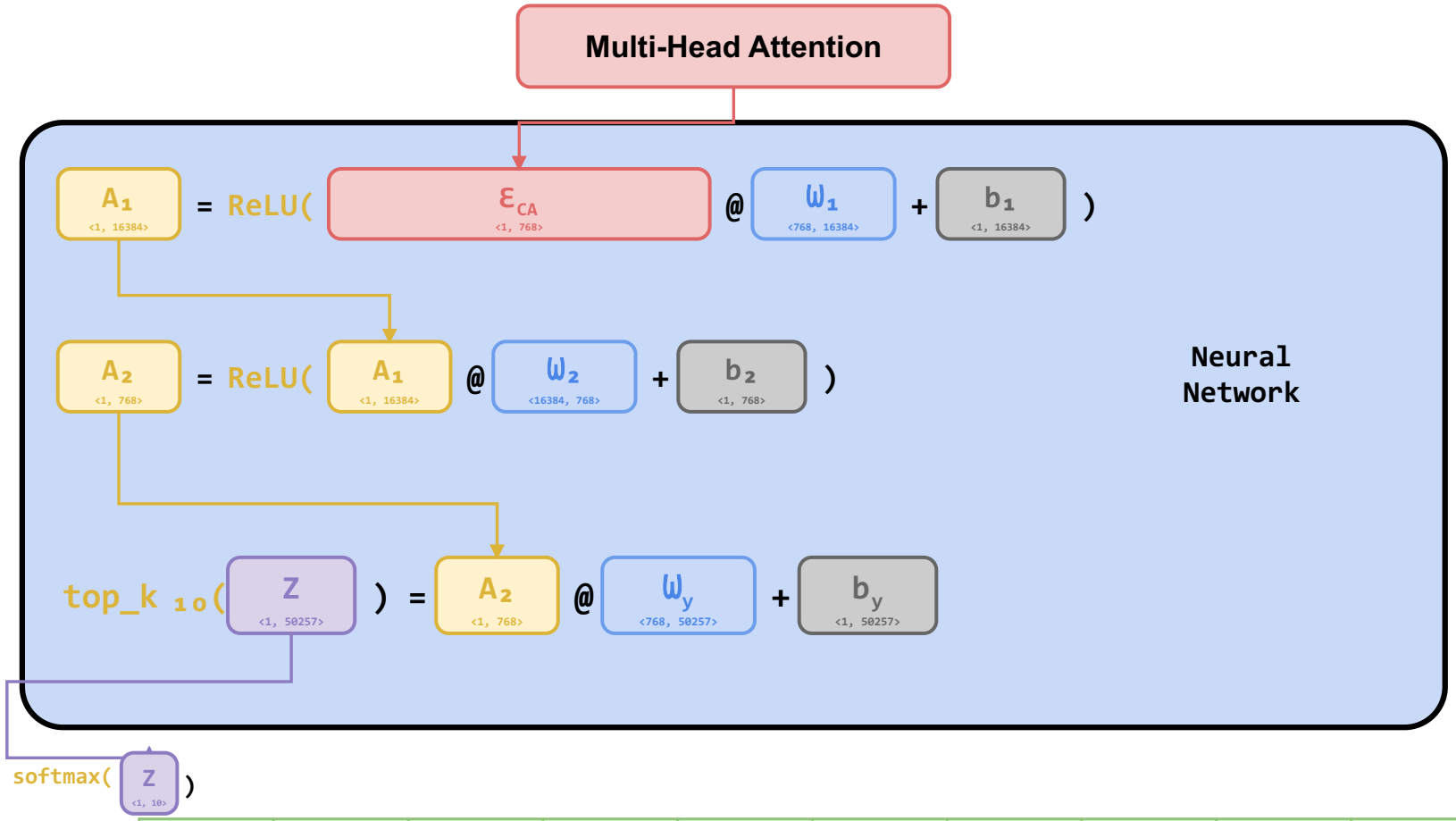
Producing the output prediction



softmax(Z <1, 10>)

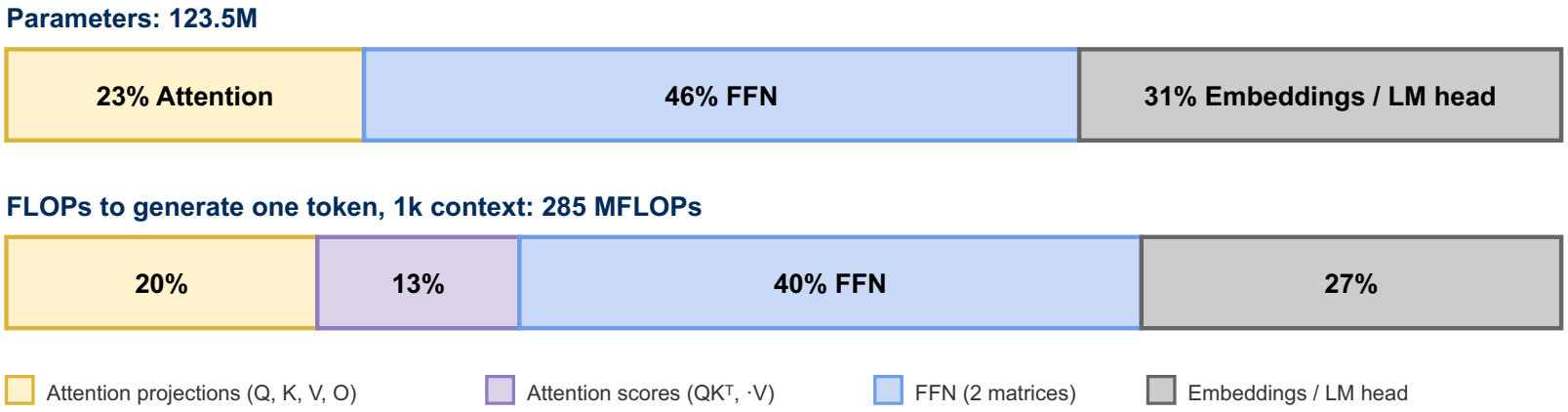
	0.1739	0.1079	0.0981	0.0779	0.0629	0.0614	0.0393	0.0241	0.0223	0.0116
Token ID	262	257	465	13	607	663	11	340	617	319
Token	" the"	"a"	" his"	"."	" her"	" its"	","	" it"	" some"	" on"

Producing the output prediction



	0.1739	0.1079	0.0981	0.0779	0.0629	0.0614	0.0393	0.0241	0.0223	0.0116
Token ID	262	257	465	13	607	663	11	340	617	319
Token	" the"	"a"	" his"	"."	" her"	" its"	","	" it"	" some"	" on"

Parameter and Compute Distribution in LLMs

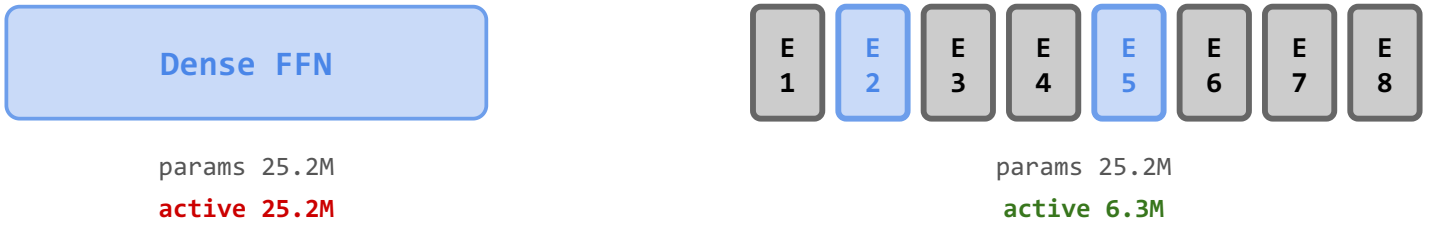


The majority of our **parameters** and **compute** in an **LLM** are actually in the **FFN (MLP)**, not the **attention block**.

To reduce this, we can use a technique called **Mixture of Experts** for the **FFN**.

Mixture of Experts

Mixture of Experts splits our **neural network** into smaller pieces called **"Experts"** of which we **only activate a few at a time** for each token.

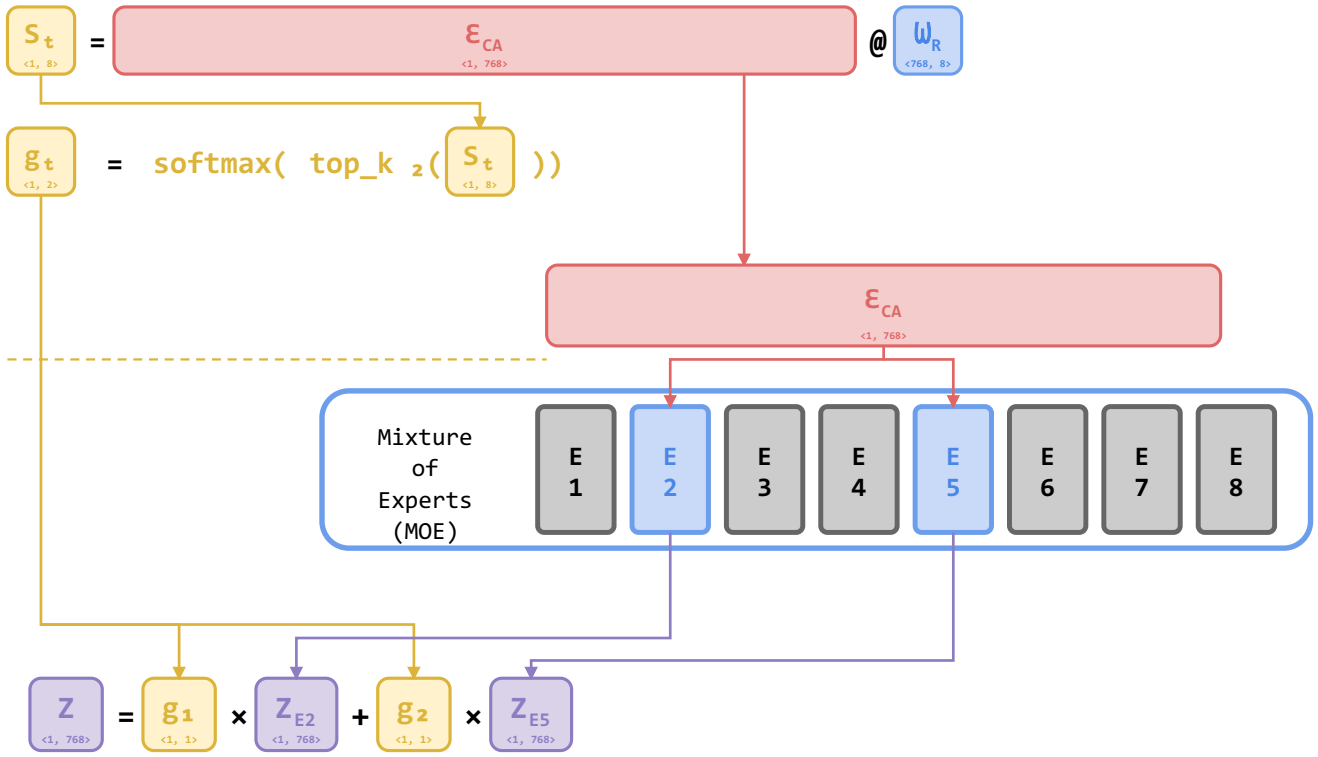


Our collection of **experts** has the **same total parameter capacity**, but fewer "active" params that we need to use during **inference**.

The number of experts is something we pick, similar to the number of attention heads.

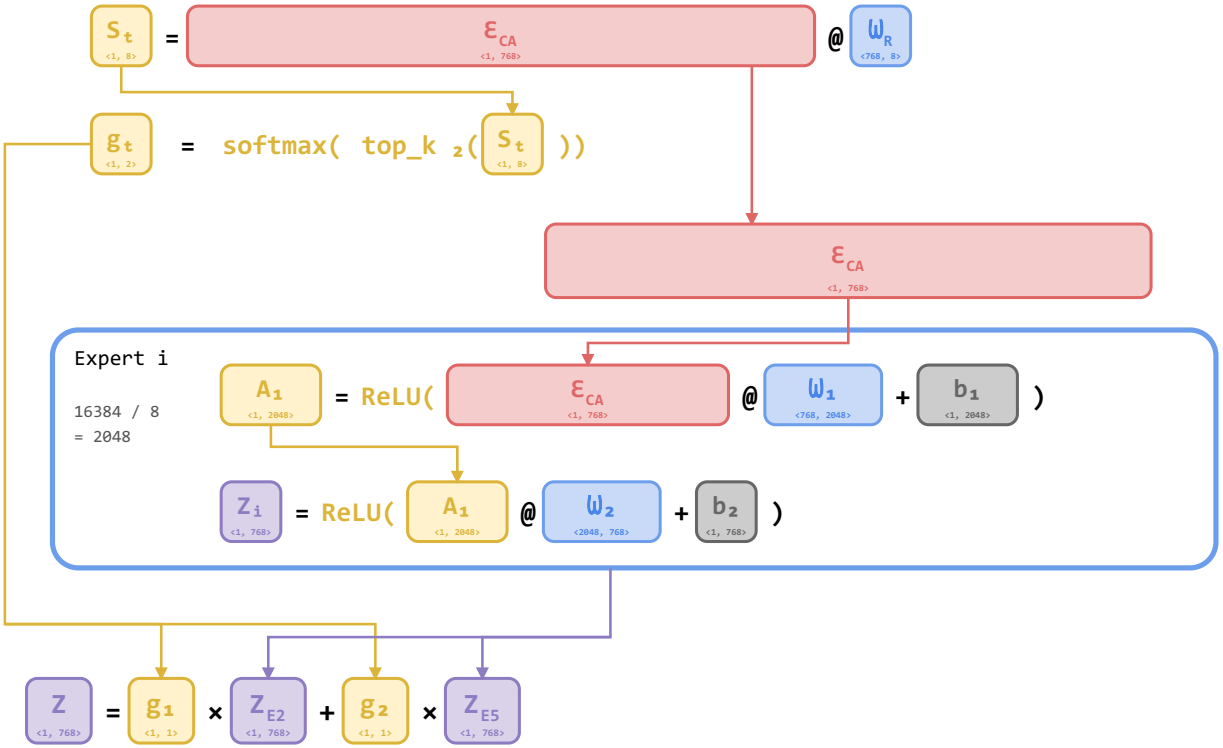
Mixture of Experts

We use W_R to create **"routing scores"** for our experts based on the attention output for the token.



If we were asking an **LLM** a question about football, we could only activate the football **expert**. This is just the intuition.

Inside an Expert

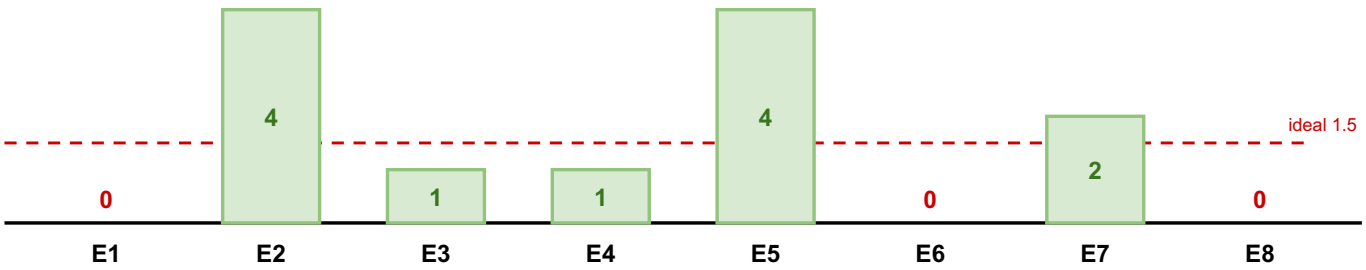


Each expert is just a smaller **FFN** but the **architecture** is identical to the original large one. This is just the intuition.

Issues with Routing to our Experts

	E1	E2	E3	E4	E5	E6	E7	E8	gates after renormalizing
"The"	0.05	**0.46**	0.03	0.04	**0.31**	0.06	0.02	0.03	E2 0.60 · E5 0.40
"cat"	0.04	**0.41**	0.05	0.06	0.08	0.03	**0.29**	0.04	E2 0.59 · E7 0.41
"sat"	0.03	**0.28**	0.02	0.05	**0.44**	0.04	0.09	0.05	E2 0.39 · E5 0.61
"on"	0.06	**0.38**	0.07	**0.33**	0.05	0.03	0.04	0.04	E2 0.54 · E4 0.46
"the"	0.04	0.07	**0.30**	0.06	**0.36**	0.05	0.08	0.04	E3 0.45 · E5 0.55
"mat"	0.03	0.05	0.04	0.06	**0.27**	0.04	**0.47**	0.04	E5 0.36 · E7 0.64

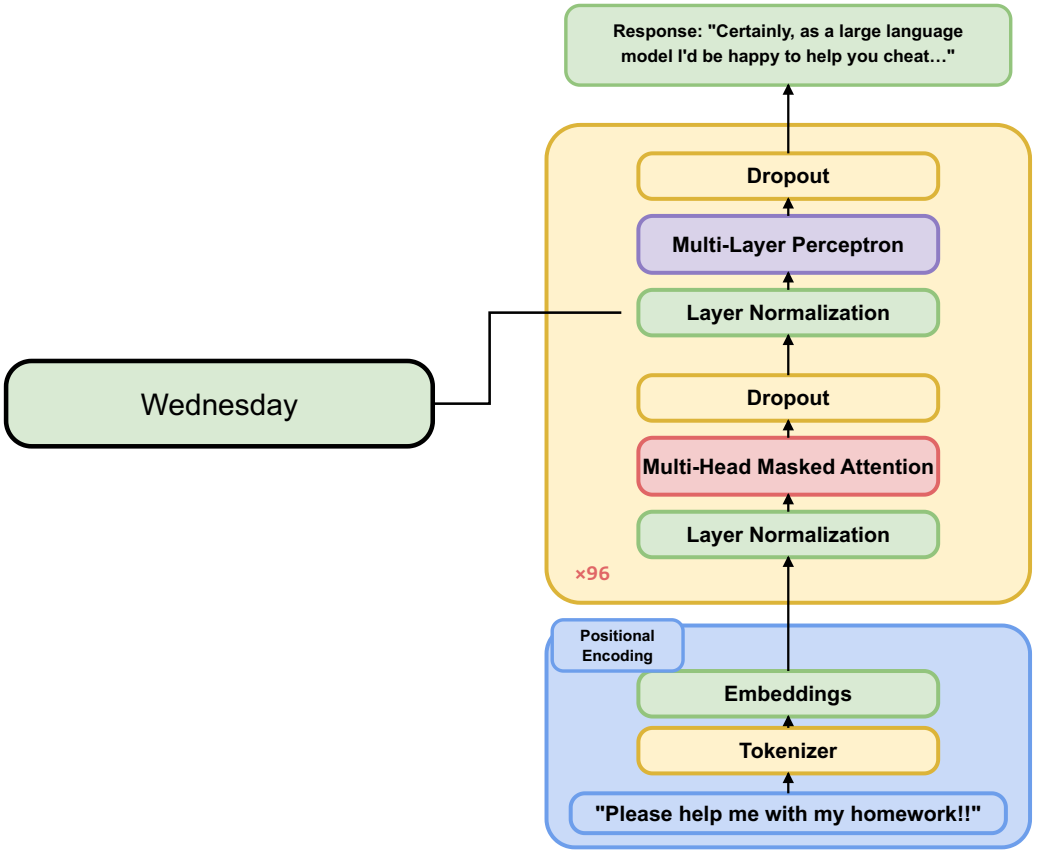
Count the gold cells in each column: tokens received per expert



Experts 2 and 5 are **overloaded**, experts 1, 6 and 8 get nothing. The **balance loss** pushes the bars toward the dashed line.

Optional capacity limit: tokens beyond an expert's cap are dropped or sent to their next choice.

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On Wednesday we'll discuss the three remaining pieces of the transformer:

- Positional Encoding
- Layer Normalization
- Dropout

After that, we'll have everything we need in order to write GPT2 from scratch!!

On that note, next week, should we do it:

A. For real (raw python files)
   a. More setup
   b. You'll have a real LLM codebase you could show off

B. Simplified (colab)
   a. Easier
   b. Less of a sense of actually writing it from scratch

Guest AMA

Zach Carmichael (Meta) & **Abby Swenor** (Fish & Richardson)
Friday, 10/09, 11:30 - 12:20, Debart 214





Building **optimization** and **observability** tooling for **agentic systems** @ **Meta**.
Technology Specialist for **Patent Law** @ **Fish & Richardson**.